

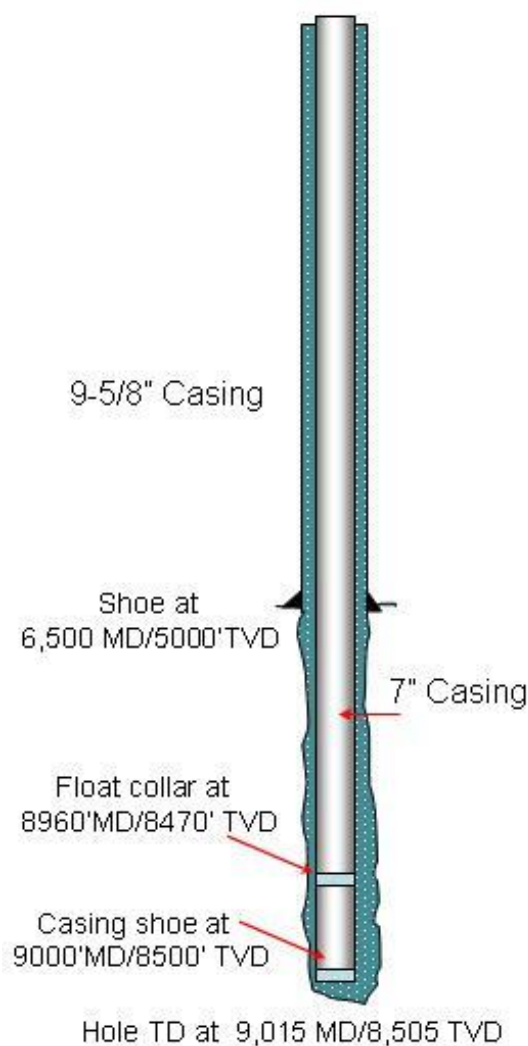
## Mid Exam

Name:

ID :

1. Explain five (5) differences between petroleum systems and geothermal systems based on the following components: source, reservoir, cap (seal), fluid, and pressure.
2. Identify and explain the five (5) major drilling systems. For each drilling system, provide one (1) example of the equipment used.

Use below diagram for **Questions 3 and 4**:



### 3. Well information:

Formation gradient 0.578 psi/ft.

- a. If you intend to use MW = 10.9 ppg, will you be overbalanced?

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- b. What is the minimum MW expected to keep this well from underbalanced?

### 4. Well information:

- Well is drilled with 8.5 inch bit to 9,015 MD/8,505 TVD with oil based mud. Estimated hole sized based on an open hole log is 8.6 inch.
- Previous casing size is 9-5/8" and its shoe is at 6,500 MD/5,000'TVD.
- Plan to run 7" casing and planned shoe depth at 9,000'MD/8,500'TVD.
- There is one float collar at 8,960'MD/8,470' TVD.
- Top of cement is 2,000 ft above 9-5/8" casing.
- Spacer volume is from top of cement to surface.

### Casing information

- 7" casing ID = 6.185"
- 9-5/8" casing ID = 8.85"

Use hole size measured from the open hole log to determine cement volume.

Determine the following items:

- a. Cement volume
  - b. Spacer volume
  - c. Displace volume
5. In a trajectory design, J-Type Well with 62 degrees inclination, Kick-Off Point (KOP) at 430 ftTVD, and average Dogleg Severity (DLS) of 2.5 degrees per 100 ft. After reaching 30 degrees inclination, hold tangent section for 720 ft, then continue building until the maximum inclination. What is the Measured Depth of End of Build?

### Usable Information

Annular capacity (between pipe and open hole) in bbl/ft =  $(D_h^2 - D_p^2) \div 1029.4$

Annular capacity (open hole) in bbl/ft =  $(D_h^2) \div 1029.4$